

ACTIVITY #1

PART A - VARIABLE IDENTIFICATION TABLE

Variable	Data	Variable	Stored
Name	Type	Type	In
name	string	instance	heap
grade	int	instance	heap
n	string	local	stack
g	int	local	stack
s1	class	local	stack
s2	class	local	stack
s3	class	local	stack

PART C - OUTPUT

John	95
Ana	90
And	95

PART B - EXPLANATION

Statement	What Happens?	Memory Impact
Student s1 = new Student ("John", 85);	Creates a new object "s1" with the name and grade "John" and "85" respectively.	1 in Stack - local reference variable (s1) 1 in Heap - new class is created (Student)
Student s2 = new Student ("Maria", 90);	Creates a new object "s2" with the name and grade "Maria" and "90" respectively.	1 in Stack - local reference variable (s2) 1 in Heap - new class is created (Student)
Student s3 = s2;	Assigns "s3" to point to the same object as "s2".	1 in Stack - local reference variable (s3) No new object is created.
s1.grade = 95;	Modifies the "grade" of the object pointed by "s1".	No memory impact. Only modifies existing heap data.

10	10
11	11
12	12

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